

KELLOG, Russian Federation

WMO: 237740

Lat: 62.4892N

Lon: 86.2814E

Elev: 198

StdP: 14.59

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -51.7	(c) -45.5	(d) -57.0	(e) 0.2	(f) -51.2	(g) -51.0	(h) 0.3	(i) -45.6	(j) 12.3	(k) 15.3	(l) 11.4	(m) 15.7	(n) 1.1	(o) 170	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.5	(c) 85.9	(d) 63.7	(e) 82.7	(f) 63.1	(g) 78.5	(h) 62.7	(i) 67.5	(j) 78.7	(k) 65.7	(l) 76.9	(m) 64.1	(n) 74.8	(o) 5.0	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.7	89.0	72.2	61.8	83.2	70.2	59.8	77.3	68.3	32.1	78.2	30.7	76.3	29.5	75.0	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	11.4	9.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	-12.7	-1.8	14.2	28.5	39.7	60.0	64.2	56.8	43.8	27.0	0.1	-5.4
	DBStd	29.14	19.10	15.86	14.79	10.16	9.18	9.54	6.78	6.73	7.88	10.54	17.69	18.61
	HDD50	9691	1945	1451	1108	644	345	30	1	20	219	714	1496	1718
	HDD65	14237	2410	1871	1573	1094	786	199	99	263	635	1178	1946	2183
	CDD50	1062	0	0	0	0	25	331	441	230	34	1	0	0
	CDD65	133	0	0	0	0	1	50	74	8	0	0	0	0
	CDH74	1873	0	0	0	0	36	773	953	107	4	0	0	0
	CDH80	548	0	0	0	0	9	250	285	4	0	0	0	0
	WSAvg	4.1	3.4	3.7	4.6	5.1	5.4	4.2	4.1	3.5	3.8	4.2	3.8	3.7
	PrecAvg	29.0	1.9	1.5	1.7	1.9	2.3	2.7	2.8	2.9	2.7	3.4	2.9	2.4
(15) Precipitation	PrecMax	35.8	3.5	2.8	3.5	2.8	4.3	5.8	4.8	4.7	4.4	4.7	4.7	3.9
	PrecMin	18.5	0.1	0.7	0.4	0.4	1.0	1.0	0.6	0.8	0.6	0.8	1.7	1.3
	PrecStd	4.3	0.7	0.5	0.7	0.6	0.8	0.9	1.1	1.0	0.9	0.8	0.8	0.7
	PrecStd	4.3	0.7	0.5	0.7	0.6	0.8	0.9	1.1	1.0	0.9	0.8	0.8	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	32.4	44.4	56.1	79.3	90.2	89.4	80.0	73.7	52.0	33.3	30.8
		MCWB	27.1	31.0	35.2	42.2	58.3	64.3	67.5	64.7	60.0	45.6	31.9	29.5
	2%	DB	22.0	28.5	39.4	50.6	67.4	86.8	85.9	77.2	66.2	45.7	31.6	27.4
		MCWB	20.6	27.0	32.8	39.5	53.5	62.7	65.6	63.1	56.8	43.0	30.6	26.5
	5%	DB	18.9	22.9	35.7	46.0	60.6	83.3	83.3	73.9	60.3	42.8	29.0	24.1
		MCWB	17.8	21.5	31.5	37.8	48.4	61.9	64.0	62.1	53.7	40.8	27.6	23.3
	10%	DB	13.8	17.8	33.0	42.1	54.4	78.9	79.8	70.2	56.1	39.3	26.6	19.4
		MCWB	12.8	16.5	30.2	35.7	45.4	61.5	63.6	59.9	50.9	37.1	25.4	18.3
	0.4%	WB	26.9	31.5	36.0	43.8	59.1	67.7	71.5	68.5	60.8	47.0	32.4	29.9
		MCDB	27.7	32.6	40.0	51.7	75.3	80.4	85.3	77.1	70.5	52.0	33.0	31.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.0	27.0	33.8	40.6	55.1	65.6	68.9	65.1	57.5	43.5	30.9	26.5
		MCDB	22.1	28.6	37.7	48.5	65.5	80.4	80.9	72.8	64.6	45.1	31.7	27.1
	5%	WB	17.7	21.4	32.1	38.5	50.6	64.2	66.9	63.0	54.4	40.8	28.1	23.4
		MCDB	18.9	22.7	34.8	44.9	58.0	77.9	77.9	71.0	59.9	42.9	29.0	24.2
	10%	WB	12.8	16.7	30.0	36.5	46.7	62.5	65.3	61.0	51.4	37.6	25.3	18.6
		MCDB	13.6	17.8	33.3	41.6	53.7	75.1	75.7	68.9	55.6	39.0	26.4	19.5
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	19.1	21.2	18.0	16.7	22.6	20.5	18.8	13.9	10.4	16.8	15.8
		MCDBR	14.1	16.4	20.0	24.1	28.1	32.1	28.3	26.1	21.3	12.7	12.8	13.3
		MCWBR	13.6	15.8	14.3	15.1	14.4	12.3	11.4	13.6	12.0	10.1	12.4	13.2
		MCDBR	13.8	17.0	16.4	20.2	24.6	25.2	23.1	22.4	19.5	12.9	13.2	13.2
	5% WB	MCWBR	13.3	16.4	12.5	13.5	14.2	11.2	11.2	12.5	11.4	10.6	12.9	13.1
		MCWBR	13.3	16.4	12.5	13.5	14.2	11.2	11.2	12.5	11.4	10.6	12.9	13.1
(40) Clear-Sky Solar Irradiance	taub		0.247	0.291	0.334	0.356	0.371	0.390	0.444	0.409	0.350	0.312	0.240	0.229
	taud		1.919	2.006	1.987	2.048	2.267	2.362	2.182	2.260	2.401	2.255	1.969	1.876
	Ebn at Noon		116	198	234	259	267	264	243	241	237	195	118	61
	Edh at Noon		12	24	36	42	37	35	40	34	24	19	11	7
	RadAvg		90	360	838	1380	1604	1774	1570	1084	646	262	113	41
(44) All-Sky Solar Radiation	RadStd		10	29	60	154	190	142	171	132	88	38	23	4

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page